

Composition Pair B1.02 + B1.20 — The Three-Level Structure Provides the Distinct Governance Scopes That Recursive Governance Applies At; Recursive Governance Provides the Coherence That Makes the Three Levels Architecturally Complete

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

Abstract

Two architectural commitments from Paper 2 stand in a co-required relationship: B1.02, which establishes the three structural levels (cell, aspect, Self), each with a distinct governance scope; and B1.20, which commits that all Paper 1 architectural commitments apply recursively at each of those levels. Neither commitment is architecturally complete without the other. B1.02 without B1.20 produces a deployment with structural level-distinctions but governance applied only at deployment scope—the Deployment-Level-Only Governance anti-pattern (B3.21). B1.20 without B1.02 produces a governance commitment that has no distinct structural levels to recurse through, collapsing recursive governance into deployment-scope governance applied uniformly to all entities. Full co-presence is required: three structurally distinct levels (B1.02), each with Paper 1 governance applied at that level’s appropriate scope (B1.20), producing a deployment governed coherently throughout at cell, aspect, and Self granularity. This note formalizes the pair: the enabling directions, the mutual dependency, the co-presence requirement, the operational integration, the detection tests, and the failure mode when co-presence breaks.

1. Pair identification

Pair: B1.02 (three structural levels—cell, aspect, Self—each with distinct functional roles and distinct governance scopes) and B1.20 (Paper 1 commitments apply recursively at each structural level—at cell scope, aspect scope, and Self scope).

Relationship type: Co-required pair. The two commitments do not stand in a sequential enabling relationship, where one is a precondition and the other a consequence. They stand in a mutual dependency: each requires the other to fulfill its architectural purpose, and the failure of either leaves

the other without the structural or governance content it needs to be more than an empty category.

Source: Paper 2 §5.2 (three architectural levels and their relational structure), §5.6 (Paper 1 inheritance at every level). The pair sits at the intersection of B2.07 (level-distinct scope), B2.14/B2.19/B2.24 (level-specific inheritance verification), and B2.110 (recursive commitments verification).

2. B1.02: the three-level structure and what it contributes

B1.02 commits that CKS deployments above single-cell scope have three named structural levels: cell, aspect, and Self. The cell is the atomic unit Paper 1 defends—a CKS artifact carrying its own substrates, orchestration rules, DNA layer, action layer, and all Paper 1 architectural commitments. The aspect is a coordination arrangement of cells serving a particular purpose, operating over its constituent cells as content domain; multiple aspects can coexist within one Self, and cells can participate in multiple aspects simultaneously through relational role membership. The Self is the integrated whole that holds multiple coexisting aspects as facets of one CKS-governed intelligence—not a bigger aspect, not a strict top-of-hierarchy, but the entity that contains the full structural composition.

What B1.02 provides, architecturally, is scope-differentiation. Cell scope and aspect scope and Self scope are not the same governance territory. A cell answers for its own substrates and orchestration rules and task-level behavior; the appropriate governance question at cell scope is whether this cell’s DNA layer, action layer, and behavior are governed correctly for the work this cell does. An aspect answers for the coordination arrangement it instantiates and the purpose it serves—the appropriate governance question at aspect scope is whether this coordination arrangement is constituted correctly, with appropriate cells, under appropriate orchestration governing their composition. A Self answers for the integrated whole—the appropriate governance question at Self scope is whether the full composition, across all aspects and cells, is coherent and correctly governed as a unit. These are distinct questions, asked at distinct granularities, producing distinct governance records.

B1.02 also commits that these levels are structurally distinguishable—not merely nominal labels applied to an undifferentiated population of entities. Level-distinguishability is testable per B2.09. Without distinguishable levels, B1.20 has no structural substrate to recurse through, and the governance architecture that both commitments together establish cannot materialize.

3. B1.20: recursive Paper 1 commitments and what they contribute

B1.20 commits that all six Paper 1 architectural commitments—human governance (authority not labor), substrate as source of truth, conflict preservation as first-class object, AI as substrate mediator, tool-agnosticism, and linear-cost composition—apply at each structural level established by B1.02. Not only at the deployment level. Not only at the cell level because Paper 1 operated at cell scope. At every level: cell scope, aspect scope, and Self scope.

This recursion is architecturally substantive rather than a formality. Paper 1’s conflict preservation, for instance, when inherited at aspect scope, means aspects themselves carry coordination conflicts as

first-class substrate state—aspects do not silently resolve conflicts that arise from cell composition. When inherited at Self scope, it means the Self carries conflicts across aspects, and Paper 1’s two-level conflict-handling structure extends naturally into the multi-level architecture. Human governance at aspect scope means humans retain the right to inspect, modify, and override aspect-level orchestration and composition structure—not merely the right to inspect individual cell behavior. At Self scope it means unified human governance over the full composition, which is the commitment that distinguishes a CKS Self from an integration-glued collection of subsystems.

What B1.20 provides, architecturally, is governance content at each scope. B1.02 names the scopes; B1.20 specifies what governing each scope means and commits the deployment to providing it. Without B1.20, the three levels exist as structural categories, but the governance questions appropriate to each level—the questions that make level-distinction architecturally purposeful—go unanswered. The levels are present but hollow.

4. Why B1.02 requires B1.20: governance completeness

B1.02 without B1.20 produces a deployment with structural levels that operate without level-appropriate governance. The cell level receives Paper 1 governance because Paper 1 commits to cell-level governance and Paper 2 preserves Paper 1 at the cell level. But aspect-level governance is absent. Self-level governance is absent. The deployment receives governance at deployment scope—an aggregate assessment that the deployment as a whole satisfies architectural requirements—without the capacity to demonstrate compliance at the level of individual cells or individual aspects operating as governed structures in their own right.

This is the Deployment-Level-Only Governance anti-pattern (B3.21). Three structural levels exist. Governance tests pass only at the deployment aggregate. Individual cells and aspects may be well-formed structurally, but their independent governance standing—their capacity to be assessed as individually governed entities with level-appropriate compliance records—is absent. The architectural purpose of having distinct levels was to enable level-appropriate governance; without B1.20, that purpose is unfulfilled, and the level distinctions are structural ornamentation rather than governance architecture.

The asymmetry with the cell level is important: Paper 1 already secures governance at cell scope. What B1.20 adds above Paper 1 is governance at aspect scope and Self scope. B1.02 defines those scopes as architecturally real; B1.20 makes them architecturally governed. Without B1.20, B1.02 adds two scope-categories—aspect and Self—that have structural presence but no independent governance standing. The deployment is governed at cell scope (via Paper 1 inheritance) and at deployment aggregate scope (via whatever top-level architecture review applies), but the intermediate and upper structural levels introduced by B1.02 are governance voids.

5. Why B1.20 requires B1.02: distinct scopes to recurse through

B1.20 without B1.02 produces a governance commitment with no structural substrate to be recursive across. Recursive governance means Paper 1 commitments apply at each structural level's scope, with that level's scope governing what the commitment means at that level. But if B1.02 is absent—if there is no three-level structure, only a flat population of entities at a single structural level—then “each structural level” names only one level. Governance at “cell scope” with no distinct aspect or Self scope above it is simply deployment-scope governance applied uniformly to all entities. The recursion through levels that B1.20 commits to has nowhere to recurse.

Even the stronger failure is available: without B1.02's structural levels, governance cannot differentiate between cell-scope questions and aspect-scope questions, because there is no architectural distinction between cells and aspects. The distinct governance vocabulary B1.20 enables—this is a cell, governed at cell scope; this is an aspect, governed at aspect scope with questions appropriate to coordination arrangements—cannot materialize. All entities are at the same level, receiving the same governance at the same granularity. Recursive governance collapses to uniform deployment-scope governance. B1.20's commitment retains the words but loses the architecture.

The co-required nature of the pair follows directly: B1.20 commits that governance is recursive through B1.02's levels; B1.02 commits that those levels exist and are structurally distinguishable. Each commitment makes the other's architectural content real. Neither can fulfill its purpose without the other present.

6. Co-presence requirement

B1.02 without B1.20 (three levels, no recursive governance): The deployment has structural levels—cells are identified as cells, aspects as aspects, Self as Self—but governance applies only at deployment scope. Individual cells are not independently governed at cell-appropriate scope. Aspects are not independently governed at aspect-appropriate scope. The Self receives deployment-scope governance, but not as a governed whole composed of governed aspects composed of governed cells. B3.21 Deployment-Level-Only Governance is the operative anti-pattern. Compliance can be demonstrated only in aggregate; the deployment cannot produce level-specific governance records because level-specific governance was never applied.

B1.20 without B1.02 (recursive governance commitment, no distinct levels): Governance attempts to apply recursively but the structural levels it would recurse through are absent. In the limiting case—no three-level structure at all—B1.20 reduces to deployment-scope governance applied uniformly. In the intermediate case—some structural hierarchy present but without B1.02's committed level-distinctions—governance may apply at some aggregation levels, but the specific cell/aspect/Self scope-differentiation B1.20 commits to is unavailable. What the deployment produces is not recursive governance in the CKS sense; it is layered governance without the level-distinguishability requirements that make the layers architecturally coherent.

Full co-presence (B1.02 and B1.20 both present): Three structurally distinct levels are established, with level-distinguishability confirmed per B2.09. Paper 1 commitments are applied at each level's scope—at cell scope through cell-level DNA and action layer governance, at aspect scope through aspect-level coordination and composition governance, at Self scope through unified governance of the full composition. The deployment is governed coherently throughout: entities at every level have level-appropriate governance records, governance reviews operate at the scope appropriate to each entity, and compliance can be demonstrated at cell granularity, aspect granularity, and Self granularity as well as at the deployment aggregate. The architectural purpose of B1.02's level-distinctions is fulfilled—each level distinction corresponds to a distinct governance scope, and each governance scope has Paper 1 governance applied at it.

7. Operational integration

At deployment design, the two commitments are established simultaneously and their interaction verified.

Three levels established per B1.02. Cells are identified as the atomic units carrying DNA and action layers and Paper 1 commitments. Aspects are identified as coordination arrangements of cells with purpose-defined scope. The Self is identified as the unified whole. Level-distinguishability tests per B2.09 confirm that each entity's level assignment is architecturally grounded—cell, aspect, and Self scope are structurally distinguishable, not nominal.

Recursive Paper 1 application per B1.20. For each entity at each level, Paper 1 commitments are applied at that level's scope. At cell level: human governance, substrate as source of truth, conflict preservation, AI-as-substrate-mediator, tool-agnosticism, and linear-cost composition all apply at cell scope as Paper 1 specifies. At aspect level: the same six commitments apply at aspect scope, with governance operating over the aspect's coordination arrangement and composition structure. At Self level: the commitments apply at Self scope, with unified human governance over the full composition.

Level-specific inheritance verification per B2.14/B2.19/B2.24. These verification points test whether Paper 1 commitments are correctly applied at each level's scope. B2.14 tests cell-level Paper 1 inheritance—confirming that each cell carries all six Paper 1 commitments as Paper 1 specifies. B2.19 tests aspect-level Paper 1 inheritance—confirming that aspects receive the same commitments at aspect scope, governing the coordination arrangement, not merely aggregating cell-level compliance. B2.24 tests Self-level Paper 1 inheritance—confirming that the Self receives the commitments at Self scope, with unified human governance over the full composition. All three verification points must pass. Passing B2.14 alone (cell-level Paper 1 inheritance confirmed, aspect- and Self-level governance absent) is the exact failure mode B3.21 names.

Recursive commitments verification per B2.110. B2.110 tests the complete B1.02 + B1.20 pair: are Paper 1 commitments applied recursively at each structural level, for all entities in the deployment? This is the pair-completeness test. B2.109 supplies the per-level governance review tests (cell-level

tests for cell governance reviews, aspect-level tests for aspect governance reviews, Self-level tests for Self governance reviews) that B2.110 aggregates across the deployment. A deployment passes B2.110 only when all entities at all levels have level-appropriate governance records and B2.14/B2.19/B2.24 have passed for each entity.

Governance reviews operate at level-appropriate scope. Because B1.02 has established distinct levels and B1.20 has established level-appropriate governance, governance reviews can proceed at the granularity appropriate to each entity. A cell governance review uses cell-level tests—does this cell’s DNA layer carry appropriate orchestration rules? Is this cell’s action layer substrate content? Are the six Paper 1 commitments satisfied at cell scope? An aspect governance review uses aspect-level tests—is this aspect’s coordination arrangement governed, with the composition of cells into the aspect carrying appropriate orchestration rules? Is conflict preservation maintained at aspect scope? A Self governance review uses Self-level tests—does unified human governance hold over the full composition? These are structurally different reviews, addressing structurally different governance questions at structurally different scopes. They become available only when the pair is fully present.

8. Detection

B2.09 (level-distinguishability tests): Are the three structural levels distinguishable? Does each entity have an assigned level that is architecturally grounded in its structural role—not a label but a structural position with distinct governance scope? This tests B1.02’s contribution to the pair. If level-distinguishability fails, B1.20 cannot apply level-appropriate governance, and the pair is absent regardless of whether B1.20 is nominally committed to.

B2.14/B2.19/B2.24 (level-specific inheritance verification): Are Paper 1 commitments applied at each level’s scope? B2.14 at the cell level, B2.19 at the aspect level, B2.24 at the Self level. Each of these tests B1.20’s contribution at one level. All three must pass. A deployment in which B2.14 passes but B2.19 and B2.24 fail has Paper 1 governance at cell scope and no Paper 1 governance at aspect and Self scope—precisely B3.21.

B2.110 (recursive commitments verification): Is the complete pair operational for all entities at all levels? B2.110 is the pair-completeness test. It passes only when B2.14/B2.19/B2.24 have passed for each entity in the deployment, and when B2.09 has confirmed level-distinguishability throughout. Passing B2.110 is the operational confirmation that B1.02 and B1.20 are both present and correctly composed.

Pair completeness: Do entities at each level have level-appropriate governance records? Can a governance reviewer request the governance record for a specific cell and receive a cell-scope record? For a specific aspect and receive an aspect-scope record? For the Self and receive a Self-scope record? If governance records exist only at the deployment aggregate, the pair is not operationally complete—B1.02 may be present but B1.20 is absent, or B1.20 is present at cell scope only.

9. Failure mode: Deployment-Level-Only Governance (B3.21)

When B1.02 is present and B1.20 is absent, the specific failure the deployment instantiates is B3.21: Deployment-Level-Only Governance. Three structural levels exist. Governance applies at deployment scope. The level distinctions B1.02 establishes are structurally real but governance-empty—they divide the deployment’s entities into categories without governing each category at its appropriate scope.

The failure is not detectable at deployment aggregate level. A deployment-scope governance review may pass: the deployment as a whole may satisfy all six Paper 1 commitments when assessed in aggregate. The failure is detectable only when the review granularity descends to individual levels—when the question asked is not “does this deployment satisfy Paper 1 commitments?” but “does this specific aspect satisfy Paper 1 commitments at aspect scope?” At that granularity, B3.21 is visible: the aspect has no governance record at aspect scope, because aspect-scope governance was never applied.

A sharper rendering of the failure: B1.02 makes three distinct governance territories available. B1.20 commits that those territories are governed. Without B1.20, the territories exist but go ungoverned; governance is applied to the deployment from outside, treating the deployment as a single territory rather than three. This collapses the architectural benefit of multi-level structure—the benefit of being able to govern each level on its own terms, at its own scope, with its own governance questions—and leaves a deployment that has multi-level structure in form but single-level governance in substance.

B3.03 (Flat Architecture anti-pattern) combined with absent B1.20 makes the failure more complete: not only is recursive governance absent, but the structural levels themselves are absent, leaving a deployment with neither the levels nor the governance B1.02 + B1.20 together provide.

10. Operational test

A deployment instantiates the B1.02 + B1.20 co-required pair if and only if all of the following hold:

- **B2.09 passes:** The deployment has three structurally distinguishable levels—cell, aspect, Self—with each entity assigned a level grounded in its structural role and distinct governance scope.
- **B2.14 passes for all cells:** Each cell carries all six Paper 1 architectural commitments applied at cell scope, as Paper 1 specifies.
- **B2.19 passes for all aspects:** Each aspect has Paper 1 commitments applied at aspect scope, governing the coordination arrangement the aspect instantiates—not merely aggregating cell-level compliance.
- **B2.24 passes for all Selves:** The Self has Paper 1 commitments applied at Self scope, with unified human governance over the full composition across all aspects.
- **B2.110 passes:** The complete pair is operational for all entities at all levels. Level-specific governance records exist at cell, aspect, and Self granularity. Governance reviews at any level can be conducted using level-appropriate tests (B2.109).

A deployment that satisfies B2.09 but fails B2.19 or B2.24 instantiates B3.21 and does not satisfy the pair. A deployment that satisfies B2.14/B2.19/B2.24 without B2.09 does not have the structural levels required for the recursive governance the pair commits to, and also does not satisfy the pair.

Source paper citation

Li, Wenxin. “The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.” April 2026. Relevant sections: §5.2 (multi-level composition and governance-boundary inheritance at every level), §5.6 (Paper 1 inheritance at every level of the three-level structure).

Li, Wenxin. “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.” April 2026. Paper 1 commitments that B1.20 commits to applying recursively: §3 (human-governed), §4 (substrate/LLM division at governance boundary), §4.2 (AI-as-substrate-mediator), §5 (conflict preservation), §6 (linear-cost composition), §7 (tool-agnosticism).

Self-citation within the series

This note extends and formalizes the following prior derivation notes in the series:

- B1.02 (three architectural levels: cell, aspect, Self with relational role membership)—one member of this composition pair
- B1.20 (Paper 1 commitments applied recursively at each structural level)—the other member of this composition pair
- B2.07 (level-distinct governance scope at each of the three levels)
- B2.09 (level-distinguishability tests as structural verification)
- B2.14, B2.19, B2.24 (level-specific Paper 1 inheritance verification at cell, aspect, and Self scope)
- B2.110 (recursive commitments verification confirming the complete B1.02 + B1.20 pair)
- B3.21 (Deployment-Level-Only Governance anti-pattern—the failure mode when B1.02 is present and B1.20 is absent)
- B4.01 (composition pair framework establishing the structure all Phase B4 notes follow)

Series note: This is derivation note B4.05 in the CKS derivation note series. Phase B4 formalizes composition pairs—architectural commitments from Paper 2 that are mutually dependent and must co-occur. B4.01–B4.04 are complete. B4.06 formalizes the B1.09 + B1.10 + B1.11 lifecycle cluster.